

BSDS: Statistics II: Introduction to Inference

First Course Handout

Instructor in charge: Minerva Mukhopadhyay (minervam@isical.ac.in)

Tutors: 1) Kolkata: Soutik Halder (haldersoutik2015@gmail.com)

2) Delhi: Sumangal Bhattacharya (sumangal.82@gmail.com)

3) Bangalore: Yasaswini B. Acharya (yashaswinibacharya@gmail.com)

Lecture hours: T 12:05 - 1:35 PM and Th 10:30 AM - 12 noon **Tutorial:** M 12:05 - 1:35 PM

Course webpage: <https://sites.google.com/view/mmukhopadhyay/home/teaching/statistics-ii-introduction-to-inference/2026?authuser=0>

Objective: This course explores key concepts in classical statistics, providing students with a comprehensive introduction to statistical inference. Topics include frequentist estimation and hypothesis testing, with an emphasis on solving problems both mathematically and through computer simulations. Real-life examples will be used to illustrate the application of statistical methods, and the effectiveness of these procedures will be evaluated through numerical simulations.

References:

1. *Introduction to Probability and Statistics for Engineers and Scientists* by Sheldon M. Ross
2. *Statistics* by D. A. Freedman, R. Pisani and R. Purves
3. *The Art of Statistics: How to Learn from Data* by D. Spiegelhalter.
4. *Probability and Statistics* by M. H. DeGroot and M. J. Schervish.
5. *Mathematical Statistics with Applications* by D. Wackerly, W. Mendenhall, R. L. Scheaffer
6. *Probability and Statistics* by Siva Athreya, Deepayan Sarkar, and Steve Tanner

Course Content: We will try to cover the following contents in this course:

1. Population vs. sample. Empirical distribution. Parametric statistical models. Need for inference on a parameter.
2. Point estimation: Concept of bias, variance, mean squared error (MSE), relative efficiency.
3. Estimation of parameters by method of moments and maximum likelihood method.
4. Sampling distributions derived from normal populations: Chi-square, t and F distributions. (Discussion of properties without derivation).
5. Introduction to asymptotic inference: Informal discussion of Weak Law of Large Numbers and Central Limit Theorem. Consistency of estimators. Approximating sampling distributions through CLT.
6. Numerical computation of sampling distributions based on independent samples from an arbitrary population. Comparison with CLT-based approximations.
7. Hypothesis testing: Null and alternative hypotheses, simple and composite hypotheses, Type I and Type II errors, level and power of a test, p-value. Neyman-Pearson Lemma, most powerful tests, unbiased tests.

8. Exact and large sample tests for binomial proportion and mean of a normal distribution (one sample case). One-sided and two-sided alternatives.
9. Exact (pooled-t) and large sample tests for the equality of two normal means. Paired t-test. Test for equality of variances of two normal populations. Test for the equality of two binomial proportions.
10. Interval estimation: Construction using pivotal quantities and critical regions. Exact and large sample confidence intervals. Confidence intervals for binomial, normal and Poisson parameters (one sample case).
11. Large sample confidence intervals for the difference of two binomial proportions. Exact and large sample confidence intervals for the difference of means of two normal populations.
12. Sample size determination in tests of hypotheses for ensuring a specified power, and in constructing confidence intervals for ensuring a specified interval width.

Grading and Evaluation:

- (i) There will be 2 **quizzes**, a mid-semester and an end-semester exams.
- (ii) Questions will be of objective and/or descriptive type.
- (iii) Marking scheme is *absolute*. The weights of different evaluations are as follows:

Quizzes: 20% on total score in quizzes (*)

Midterm: 30%

Endterm: 50%

(*) A student's total score in quizzes will be the maximum of their two quizzes (in cases three quizzes are conducted).

(**) In case there is a class project, then the weight of the same will be adjusted with the weight of the quizzes.

- (iv) If a student misses a quiz, then s/he will be evaluated based on the other quiz. If s/he misses both the quizzes, then s/he will get zero marks in quiz.
- (v) The answer scripts after evaluation will be uploaded in the GradeScope.

General Guidelines:

- (i) Students are expected to spend 10 - 12 hours per week for good performance in this course.
- (ii) If a student found to follow unfair means (in any of the exams), then irrespective of the circumstances his/her exam will be cancelled (with no supplementary exam).